

Malware Analysis Lab – Answer Sheet

Name:

Date:

Task 1 – Static Analysis (PEStudio 9.51)

Instructions: Open **hello.exe** in PEStudio and record the following:

Architecture: Executable, 64 bit, GUI

Entry Point (EP):entry-point > first 32 bytes (hex),F9 99 77 79 00 00 89 9F 79 9F 00 F7 00 00 79 97 89 80 00 F8 00 00 F9 98 99 80 00 07 87 00 89 99 > location,0x00021AA0

Number of Imports: Six number of Imports

Key Imports (list 2–3)

- exit,-,implicit,-,0x00000000,0x00000000,msvcrt.dll
- MessageBoxA,-,implicit,-,0x00000000,0x00000000,USER32.dll
- LoadLibraryA,-,implicit,-,0x00000000,0x00000000,KERNEL32.DLL

Sections (VA, Virtual Size, Raw Size / Offset, Permissions)

- Section name- UPX0, UPX1, UPX2
- VA virtual-address (begin),0x00001000,0x0001A000,0x00022000
- virtual-size (139264 bytes),0x00019000 (102400 bytes),0x00008000 (32768 bytes),0x00001000 (4096 bytes)
- raw-size (32768 bytes),0x00000000 (0 bytes),0x00007E00 (32256 bytes),0x00000200 (512 bytes)

Task 2 – Dynamic/Hybrid Analysis (VirusTotal)

Instructions: Upload the file or SHA256 hash to VirusTotal and record observations.

Detection Ratio: 44/72

Example AV Names

File Activity

Files opened

\\crashpad_872_ROYQTOELESYLJBIY

C:\\Windows\\Fonts\\staticcache.dat

C:\\Windows\\system32\\rpcss.dll

C:\\Users\\user\\Desktop\\hello.exe

C:\\Windows\\AppPatch\\sysmain.sdb

C:\\Windows\\Globalization\\Sorting\\sortdefault.nls

C:\\Windows\\SYSTEM32\\CRYPTBASE.DLL

C:\\Windows\\SYSTEM32\\CoreMessaging.dll

C:\\Windows\\SYSTEM32\\CoreUIComponents.dll

C:\\Windows\\SYSTEM32\\TextShaping.dll

Files dropped

C:\\Program Files (x86)\\Google\\GoogleUpdater\\141.0.7376.0

C:\\Program Files (x86)\\Google\\GoogleUpdater\\141.0.7376.0\\Crashpad

C:\\Program Files

(x86)\\Google\\GoogleUpdater\\141.0.7376.0\\Crashpad\\attachments

C:\\Program Files (x86)\\Google\\GoogleUpdater\\141.0.7376.0\\Crashpad\\metadata

C:\\Program Files (x86)\\Google\\GoogleUpdater\\141.0.7376.0\\Crashpad\\reports

C:\\Program Files (x86)\\Google\\GoogleUpdater\\141.0.7376.0\\uninstall.cmd

C:\\Program Files (x86)\\Google\\GoogleUpdater\\141.0.7376.0\\updater.exe

C:\\Program Files\\Google872_446670514

Registry changes

Registry keys opened

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\FontLink\SystemLink

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\LanguagePack\DataStore_V1.0

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\LanguagePack\DataStore_V1.0\DataFilePath

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\LanguagePack\DataStore_V1.0\Disable

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\LanguagePack\SurrogateFallback

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\LanguagePack\SurrogateFallback\Tahoma

HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Themes\
Personalize

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Explorer\Sh
ell Folders

HKEY_LOCAL_MACHINE\OSDATA\Software\Microsoft\Windows
NT\CurrentVersion\AppCompatFlags

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\AppModel\Lookaside\machine

Network Activity

DNS Resolutions

a1666.dscr.akamai.net

assets.msn.com

assets.msn.com-ion.edgesuite.net

ecs-office.s-0005.dual-s-msedge.net

nexusrules.officeapps.live.com

IP Traffic

TCP 20.96.52.198:443

TCP 151.101.22.172:80

TCP 184.27.218.92:80

UDP 192.168.0.72:137

MITRE Tactics and Techniques

- **5 LOW**
- **4 INFO**

What tool was used to pack this software for defensive evasion? Obfuscated Files or Information was used to pack this software for defense evasion.

Task 3 – Reflection

Instructions: Summarize your findings in 3–5 sentences, comparing static and dynamic analysis.

Reflection: Static analysis examines a file without executing it, allowing analysts to quickly identify suspicious patterns, code structures, or known signatures. Dynamic analysis, on the other hand, executes the file in a controlled environment (like a sandbox) to observe its real behavior, including system changes and network activity. Using both approaches together provides a more comprehensive and reliable method for detecting potential threats, as each compensates for the limitations of the other.